AIDD8122 • FINAL PROJECT PRESENTATION

Trust Before Throughput

Benchmarking, uncertainty-aware screening and multi-objective generative design of FAP-selective radioligand warheads

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Outline

Ten to fifteen minutes, four results, one message about when to trust the model.

01 **Background**

FAP radiopharmaceuticals and what limits the current ligands

02 **Research questions**

Accuracy, transfer, trust, and generative design

03 **Methods**

Curation, three split protocols, conformal intervals, graph GA

04 **Results**

Benchmark, selectivity, uncertainty, temporal drift, design campaigns

05 **Conclusions**

Take-home messages, limitations, what I would do next

FAP is the fastest-moving small-molecule target in nuclear medicine

A stromal address with a clean selectivity profile

FAP sits on cancer-associated fibroblasts, is largely absent from healthy tissue, and accepts drug-like small molecules.

One warhead, four radionuclides

Quinoline-carboxamide FAPIs: 68Ga / 18F-AlF for imaging, 90Y / 177Lu for therapy — only the chelator changes.

Imaging contrast has not become therapeutic benefit

Tumour retention is short, responses are rare and myelosuppression is dose-limiting — each design cycle costs weeks of chemistry plus an animal study.

The design levers

Warhead · linker · chelator · albumin binder — with selectivity over PREP and DPP4 as the hard constraint.

28

tumour entities imaged

68Ga-FAPI PET/CT, Kratochwil 2019

4

radionuclides, one warhead

68Ga / 18F-AlF imaging · 90Y / 177Lu therapy

14%

objective response (4/28); DCR 46%

Fu 2025 · 21% grade 3/4 haematological tox

Four questions a radiochemist would actually ask

Published FAP QSAR models report **random-split $R^2 \approx 0.75$**. That answers an interpolation question — the four questions below are what a design decision actually depends on.

RQ1

How accurate is a FAP affinity model under real use?

Random vs scaffold vs time split: interpolation, scaffold hopping, prospective prediction.

RQ2

Does paralogue bioactivity transfer to FAP?

PREP and DPP4 have 1.3x and 11x more data; can selectivity be predicted at all?

RQ3

When are the uncertainty estimates honest?

Applicability-domain gating plus split-conformal intervals at the 80% and 90% levels.

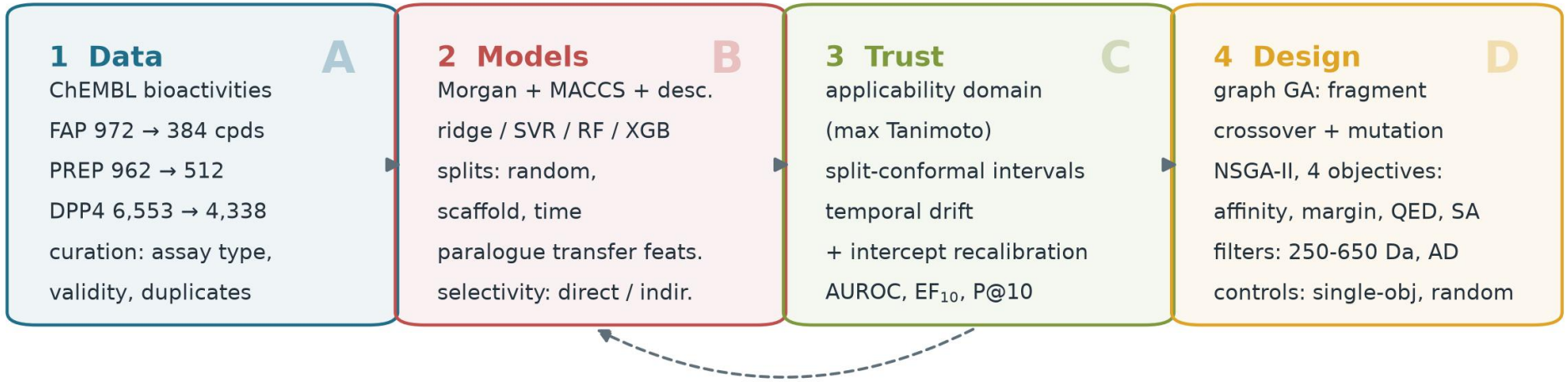
RQ4

What does multi-objective design add?

Graph GA with NSGA-II over affinity, margin, QED and SA, against two controls.

Success criteria fixed in advance: scaffold-split R^2 above 0.5 and screening AUROC above 0.85, and conformal coverage that can be predicted from the applicability domain.

Workflow: from ChEMBL records to designed warheads



Outputs

- dataset + curation log
- benchmark, 5 models × 3 splits
- calibrated error bars
- Pareto set of novel warheads

Every experiment is repeated over 5 random seeds with identical feature blocks and hyper-parameters; curated tables, figures and code are released with the report.

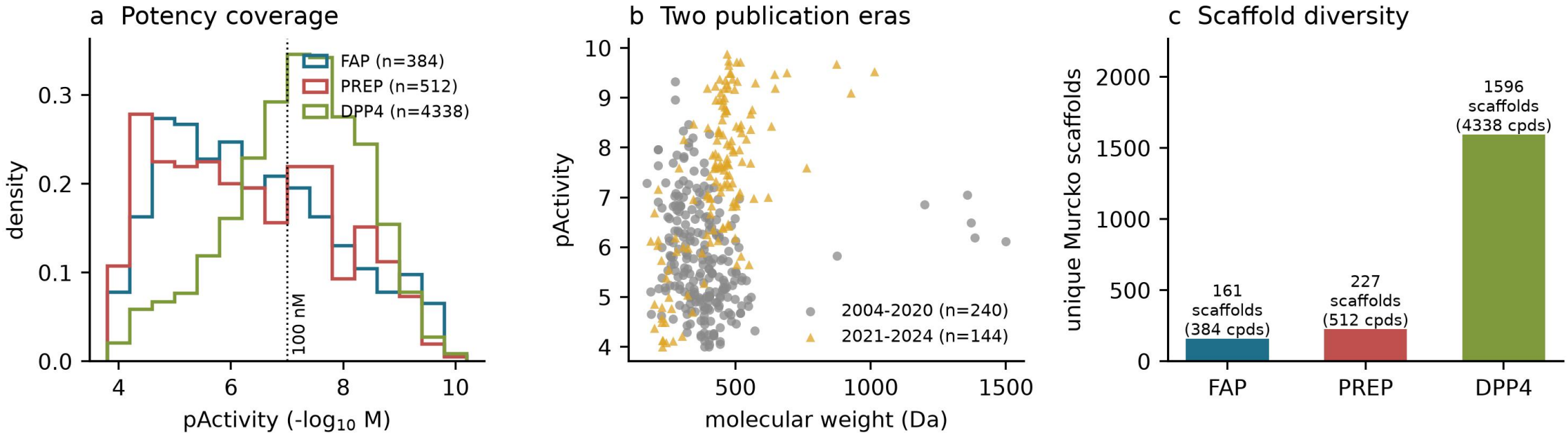
Data 972 FAP records, curated with an auditable filter log

Models Morgan + MACCS + descriptors; ridge / SVR / RF / XGBoost

Trust applicability domain + conformal intervals

Design graph GA, NSGA-II, four objectives

The public FAP record is small, noisy and non-stationary



972 → 384

unique FAP inhibitors after curation, 161 Murcko scaffolds

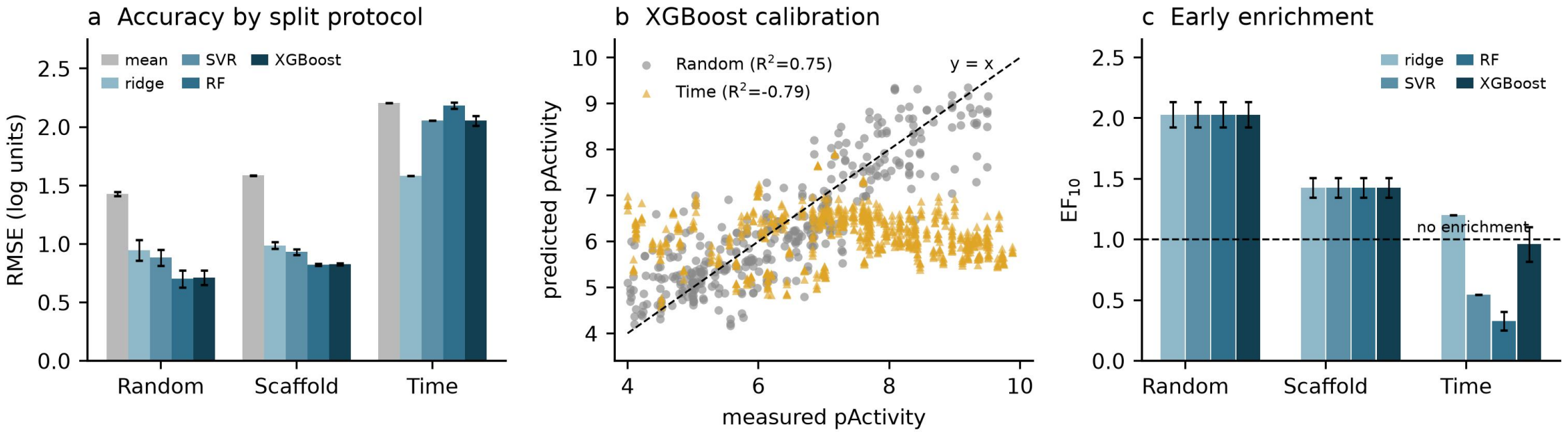
0.67 log

median replicate SD — the error floor no model can beat (best RMSE: 0.70)

+2.0 log

potency jump after 2018, with median MW +80 Da — a new chemical series

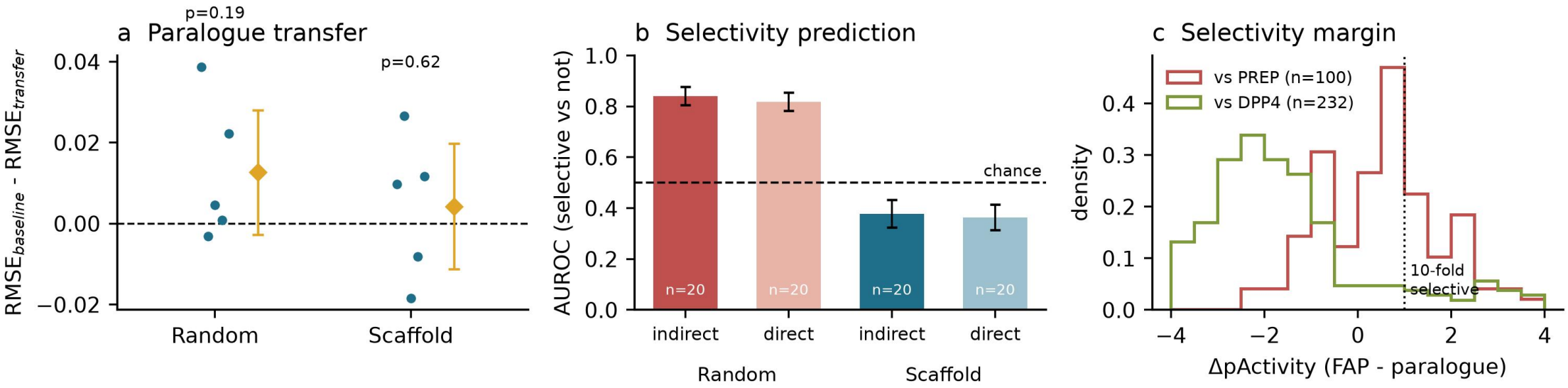
The split, not the algorithm, decides the conclusion



XGBoost R^2: **0.75** (random) → **0.68** (scaffold) → **−0.79** (time). A model trained on pre-2019 chemistry cannot rank the FAPI series at all — enrichment EF10 = 0.96, i.e. no better than random picks.

RF ≈ XGBoost, both clearly ahead of SVR and ridge on the interpolation protocols; ridge degrades least under time splitting because it relies on additive physicochemical trends rather than scaffold-specific fingerprints.

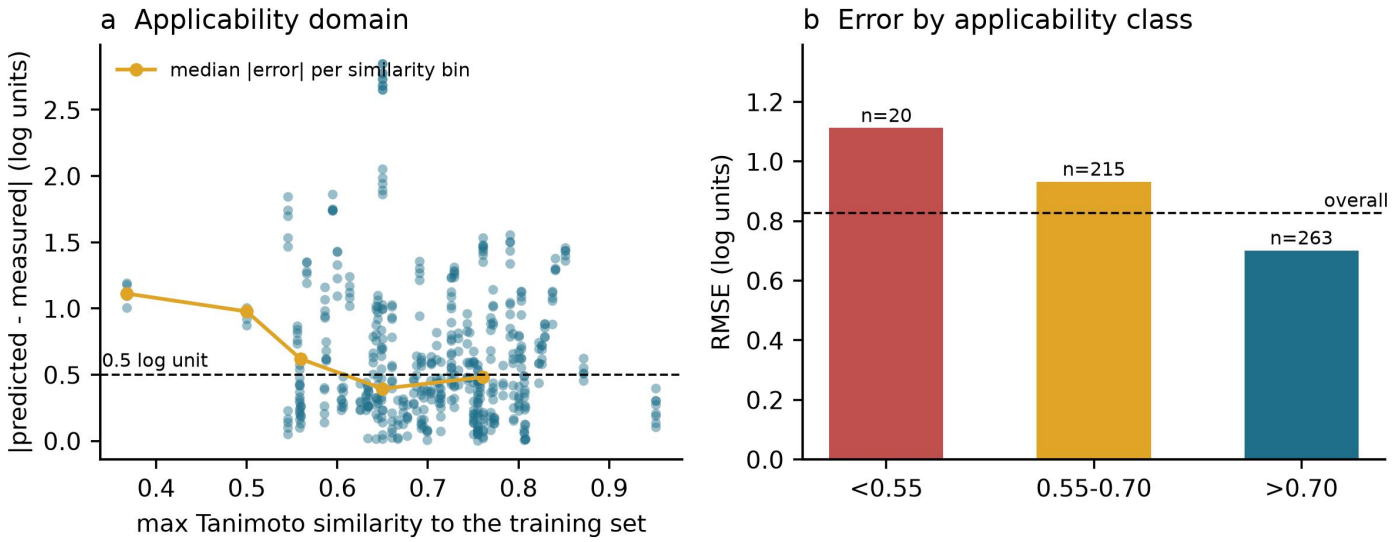
Selectivity is a cliff, not a slope



Selectivity AUROC **0.84** within known scaffolds — but **0.38 (below chance)** on new scaffolds, over 20 seeds. Used as a filter, the model would enrich the non-selective series.

Direct modelling of ΔpActivity and subtraction of two single-target models are indistinguishable. DPP4 is the dominant liability: 86% of dual-measured compounds are non-selective vs DPP4 (mean Δ = −1.75 log); the effect reverses only in the small mixed-assay stratum (n = 30, report §5.5).

Error bars you can defend — and where you cannot



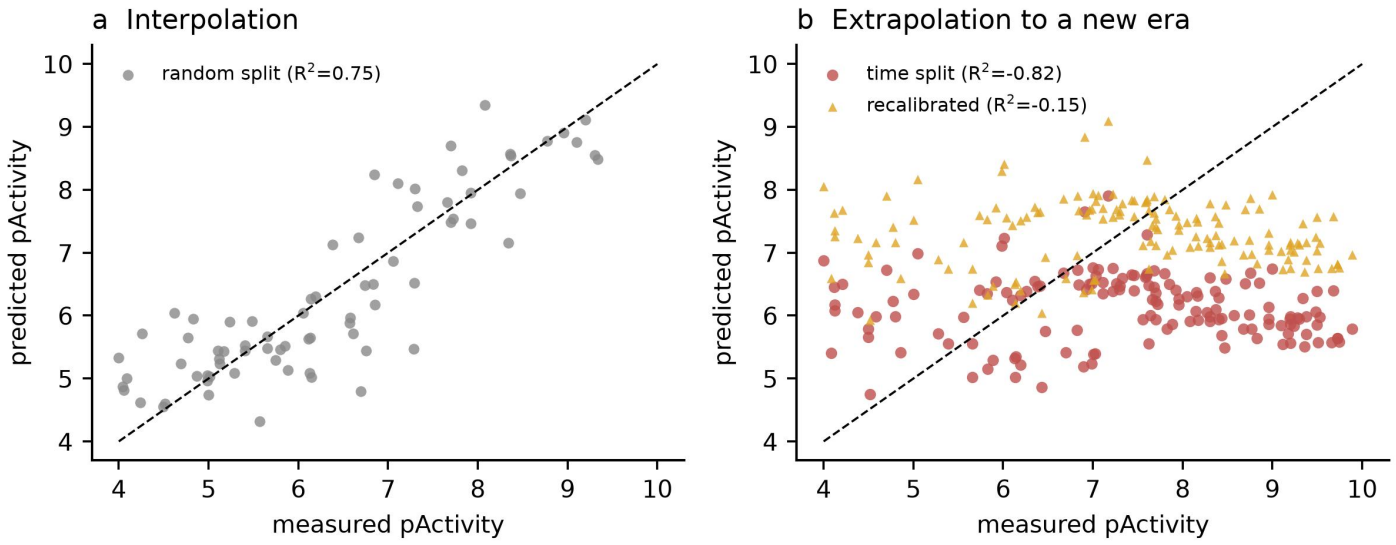
Error rises monotonically with distance from the training set: RMSE 0.70 at similarity above 0.7 versus ≈1.0 below 0.5. The rule that follows: quote a conformal interval only when the nearest-neighbour similarity exceeds 0.6 — below that, rank a series internally, but keep the interval out of the synthesis decision.

Conformal coverage (nominal 80% / 90%)

Protocol	80%	90%
Random split (control)	0.79	0.89
Scaffold split, in domain	0.77	0.88
Scaffold split, all	0.73	0.87
Scaffold split, out of domain	**0.54**	**0.77**

Exchangeability holds under random splitting (control matches theory) and fails for out-of-domain compounds — the interval is optimistic exactly where it matters most.

Prospective failure is a level shift, not a ranking signal



Train on compounds published up to 2018, test on 2019–2024. The model under-predicts every new compound by roughly an order of magnitude — it has simply never seen the quinoline FAPI series.

+1.18 log

systematic offset of the raw model on post-2018 compounds

−0.82 → −0.15

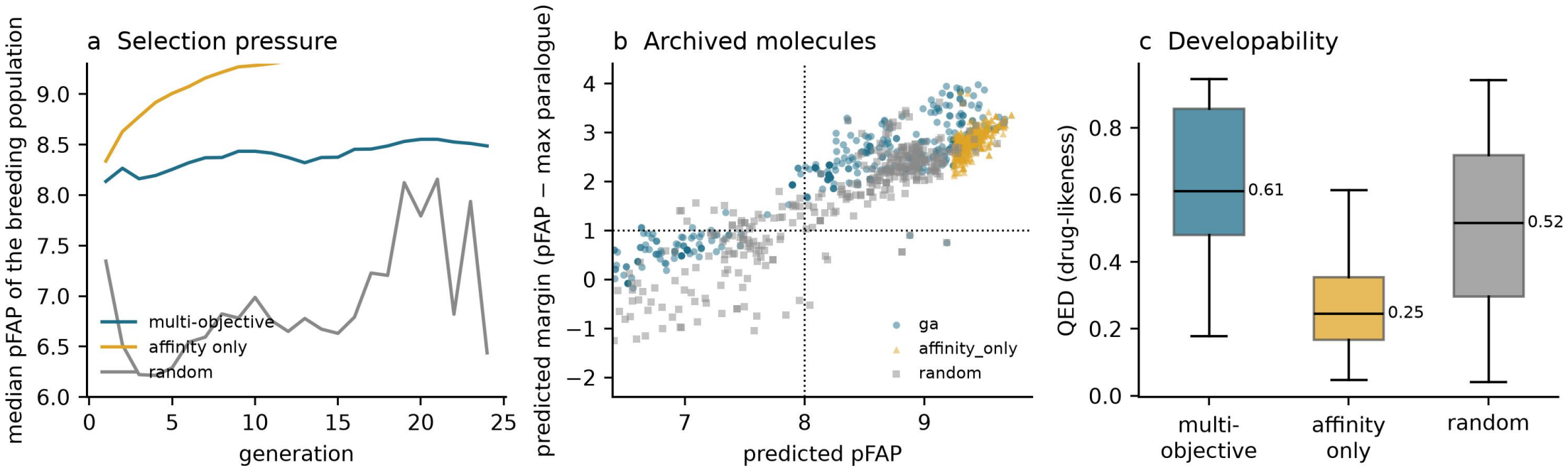
R^2 after recalibrating one intercept on 25% of the test set

−0.12

Spearman correlation — unchanged by recalibration: ranking information is absent

Recalibration restores the level, never the ranking.

Generative design: the objectives shape the chemistry



Multi-objective

115 scaffolds • QED 0.61

median MW 413 Da, SA 2.93 — the most usable chemistry

Affinity only

83 scaffolds • QED 0.25

median MW 548 Da — the model is exploited, chemistry degenerates

Random

80 scaffolds • QED 0.52

no selection pressure; population fitness drifts down

All campaigns hit the affinity-model ceiling (pFAP 9.6–9.7) in a few generations; 361 of 384 archive molecules are novel, but the top hits are warheads, not radioligands — no chelator, no leaving group (Appendix B).

Take-home messages

1

The evaluation protocol outweighs the algorithm

The same XGBoost model is excellent (R^2 0.75) or worse than the mean predictor (R^2 −0.79) depending only on the split. A claim about FAP prediction that does not specify the split is unfalsifiable.

2

Assay noise sets the ceiling, not model architecture

Median replicate SD 0.67 log units vs best RMSE 0.70. The highest-value intervention is a consistent assay campaign, not a bigger network.

3

Selectivity must be verified, not optimised in silico

AUROC 0.84 inside known scaffolds, 0.38 across scaffolds. Conformal intervals are honest only inside the applicability domain (0.79 vs 0.54 coverage at 80%).

4

The deliverable is a calibrated decision rule

Use the model above similarity 0.6, distrust it across series and eras; multi-objective design keeps the chemistry usable while single-objective greed destroys it.

Limitations — and what I would do next

Limitations

- ChEMBL snapshot: publication bias towards potent compounds; assay conditions inconsistent across ~60 papers.
- Ligand-based only — no docking or protein-structure information; physicochemical coincidence cannot be excluded.
- The generative scorer is the same model whose failure modes are documented; pFAP > 9 means "inside the confident region", not "10 nM".
- Similarity gate at 0.40 constrains novelty by construction.

Next steps

- Centralised, consistent assay campaign for a few hundred FAP inhibitors — the cheapest way to lower the error floor.
- Structure-aware features (docking poses, protein contacts) to separate sub-pocket match from coincidence.
- Prospective time-split validation: freeze the model, score everything published in the next 12 months.
- Dual-objective active learning: let the selectivity assay, not the model, decide which generated scaffold is synthesised.

Key references

[1] Lindner T, et al. Development of quinoline-based theranostic ligands for the targeting of fibroblast activation protein. J Nucl Med. 2018;59(9):1415-1422.

[2] Kratochwil C, et al. 68Ga-FAPI PET/CT: tracer uptake in 28 different kinds of cancer. J Nucl Med. 2019;60(6):801-805.

[3] Giesel FL, et al. FAPI-74 PET/CT using either 18F-AlF or cold-kit 68Ga labeling. J Nucl Med. 2021;62(2):201-207.

[4] Baum RP, et al. Feasibility, biodistribution, and preliminary dosimetry in peptide-targeted radionuclide therapy of diverse adenocarcinomas using 177Lu-FAP-2286. J Nucl Med. 2022;63(3):415-423.

[5] Fendler WP, et al. Safety and efficacy of 90Y-FAPI-46 radioligand therapy in patients with advanced sarcoma and other cancer entities. Clin Cancer Res. 2022;28(19):4346-4353.

[6] Fu H, et al. 177Lu-LNC1004 radioligand therapy in patients with end-stage metastatic cancers: a single-centre, phase II study. Clin Cancer Res. 2025;31(8):1415-1426.

[7] Mayr A, et al. Large-scale comparison of machine learning methods for drug target prediction on ChEMBL. Chem Sci. 2018;9(24):5441-5451.

[8] Wallach I, Heifets A. Most ligand-based classification benchmarks reward memorization rather than generalization. J Chem Inf Model. 2018;58(5):916-932.

[9] Sheridan RP. Time-split cross-validation as a method for estimating the goodness of prospective prediction. J Chem Inf Model. 2013;53(4):783-790.

[10] Angelopoulos AN, Bates S. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. arXiv:2107.07511. 2021.

[11] Jensen JH. A graph-based genetic algorithm and generative model / Monte Carlo tree search for the exploration of chemical space. Chem Sci. 2019;10(12):3567-3572.

[12] Deb K, et al. A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Trans Evol Comput. 2002;6(2):182-197.

Thank you

Trust before throughput: the useful output of a model on a small, drifting target is a calibrated decision rule — not a single predicted number.

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Code, curated data and filter logs: available in the submitted project repository